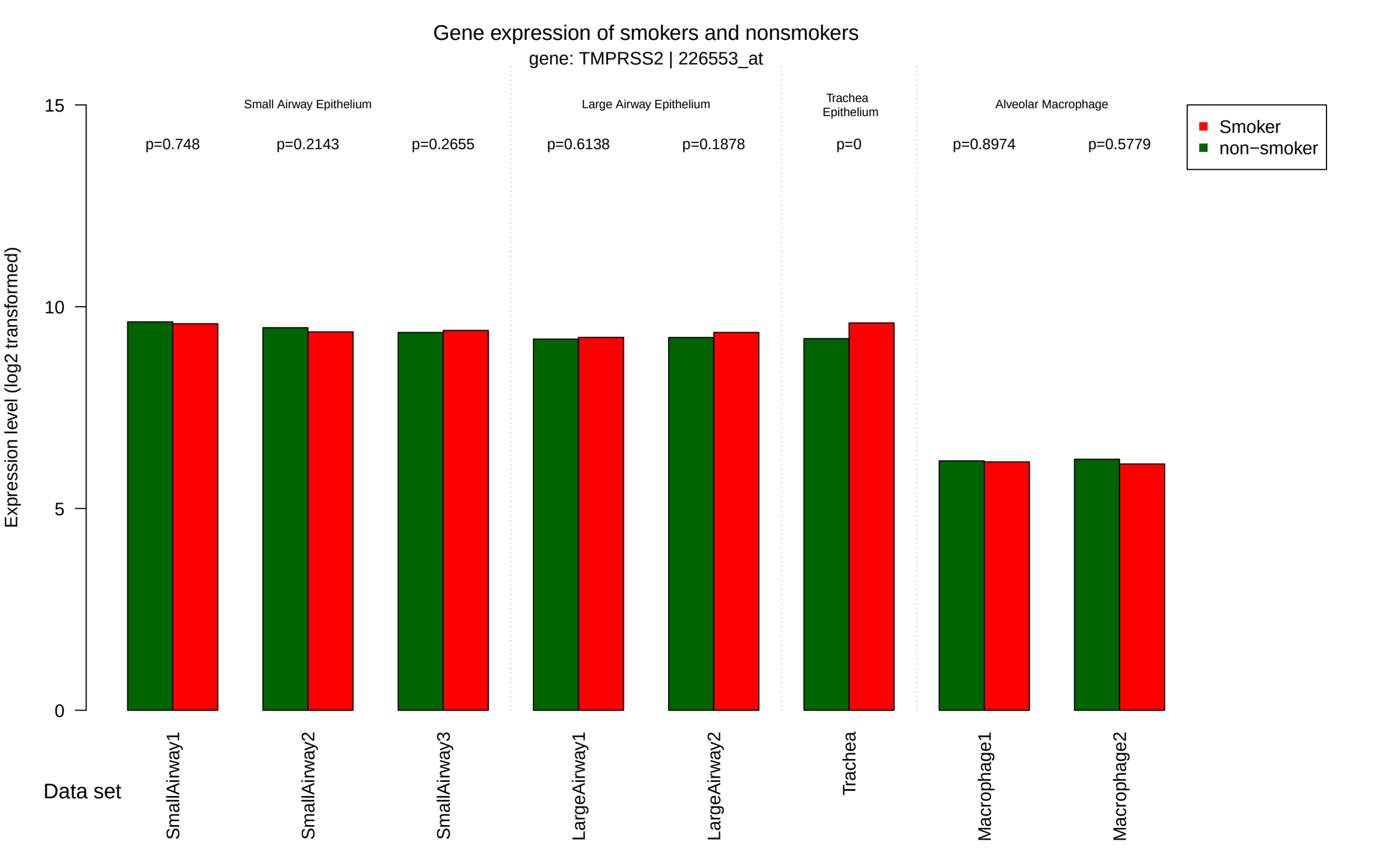
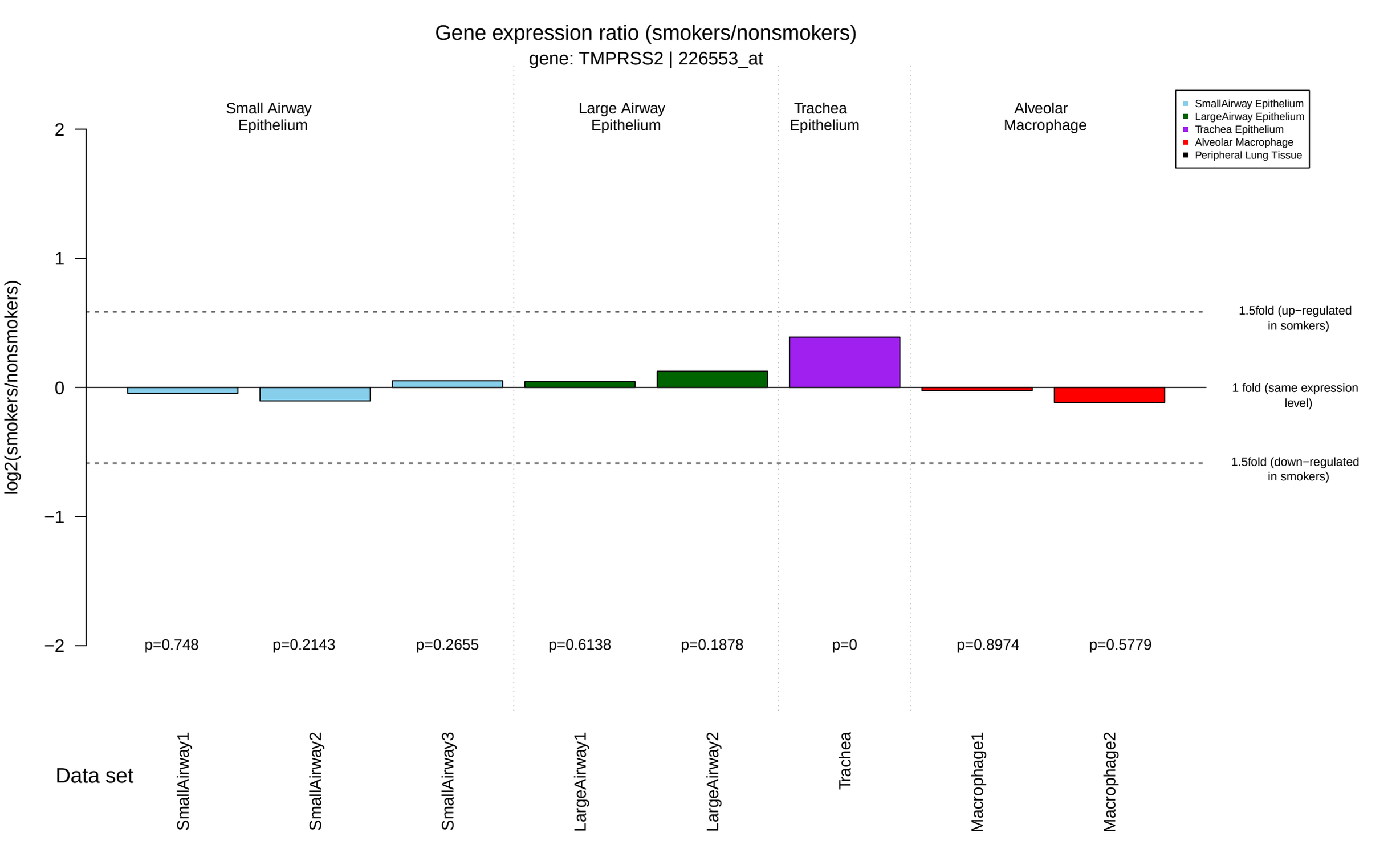
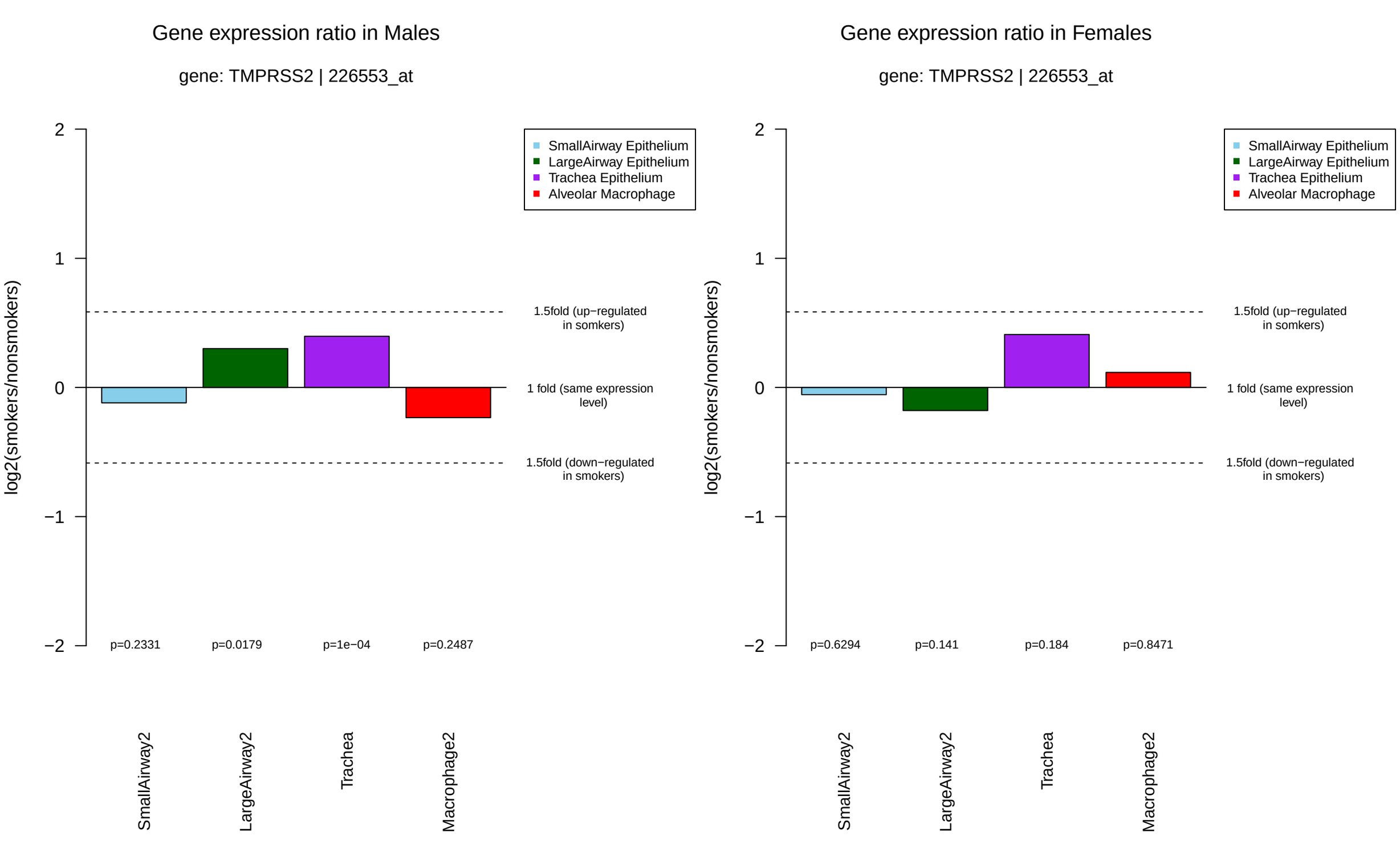
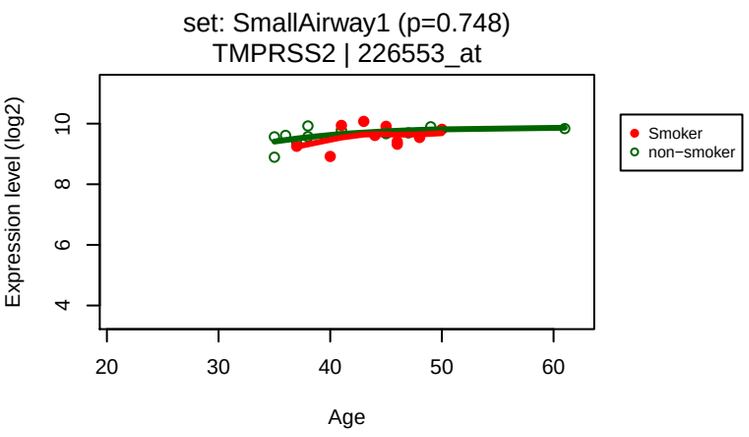
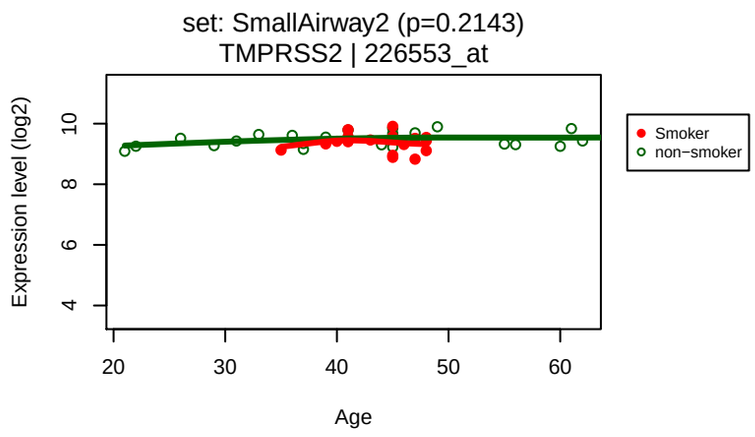
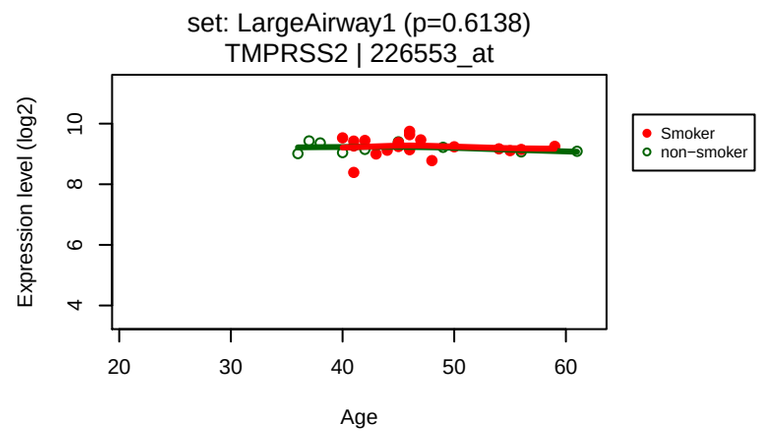

Gene expression of smokers and nonsmokers
gene: TMPRSS2 | 226553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.748
p=0.2143
p=0.2655
p=0.6138
p=0.1878
p=0
p=0.8974
p=0.5779
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

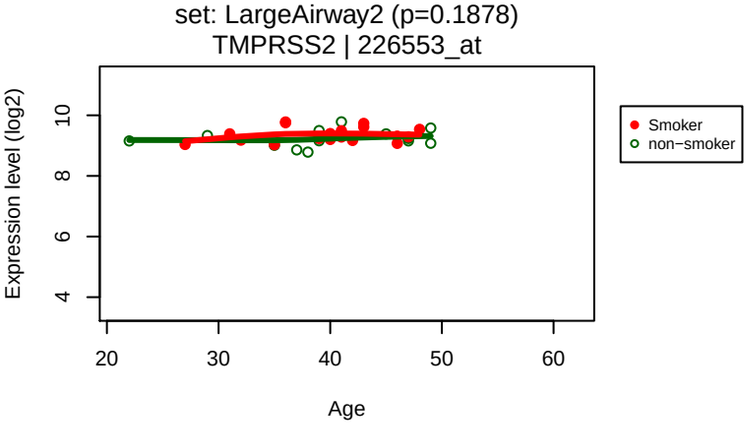

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS2 | 226553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.748
p=0.2143
p=0.2655
p=0.6138
p=0.1878
p=0
p=0.8974
p=0.5779
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

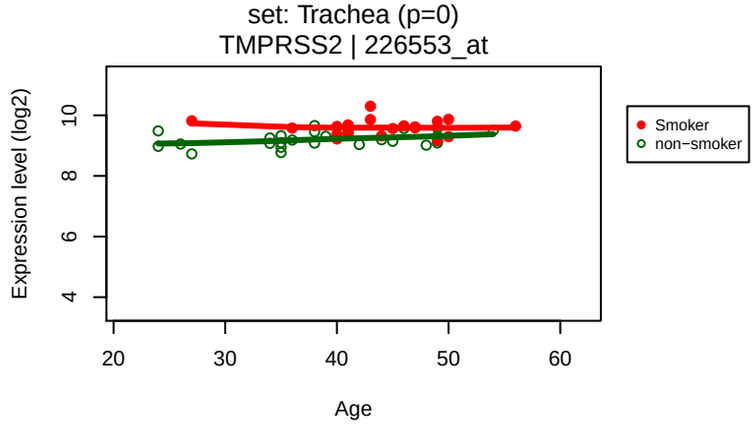

Gene expression ratio in Males
gene: TMPRSS2 | 226553_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2331
p=0.0179
p=1e−04
p=0.2487
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS2 | 226553_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6294
p=0.141
p=0.184
p=0.8471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

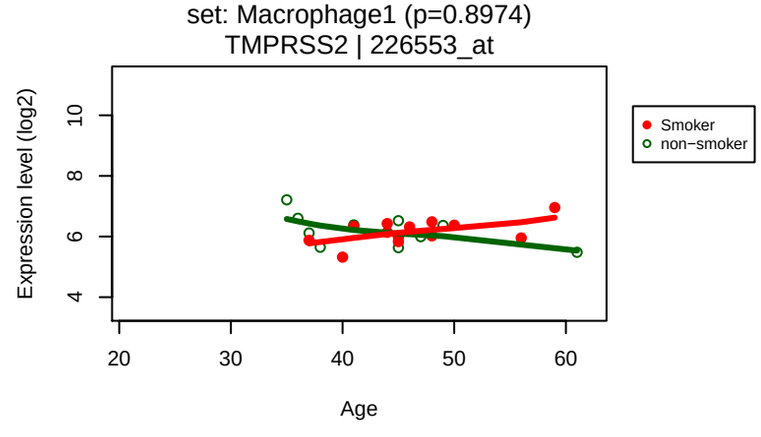

set: SmallAirway1 (p=0.748)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker

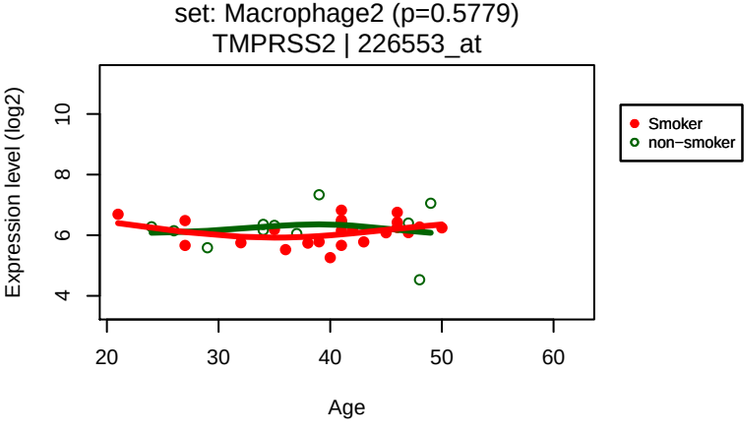

set: SmallAirway2 (p=0.2143)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker

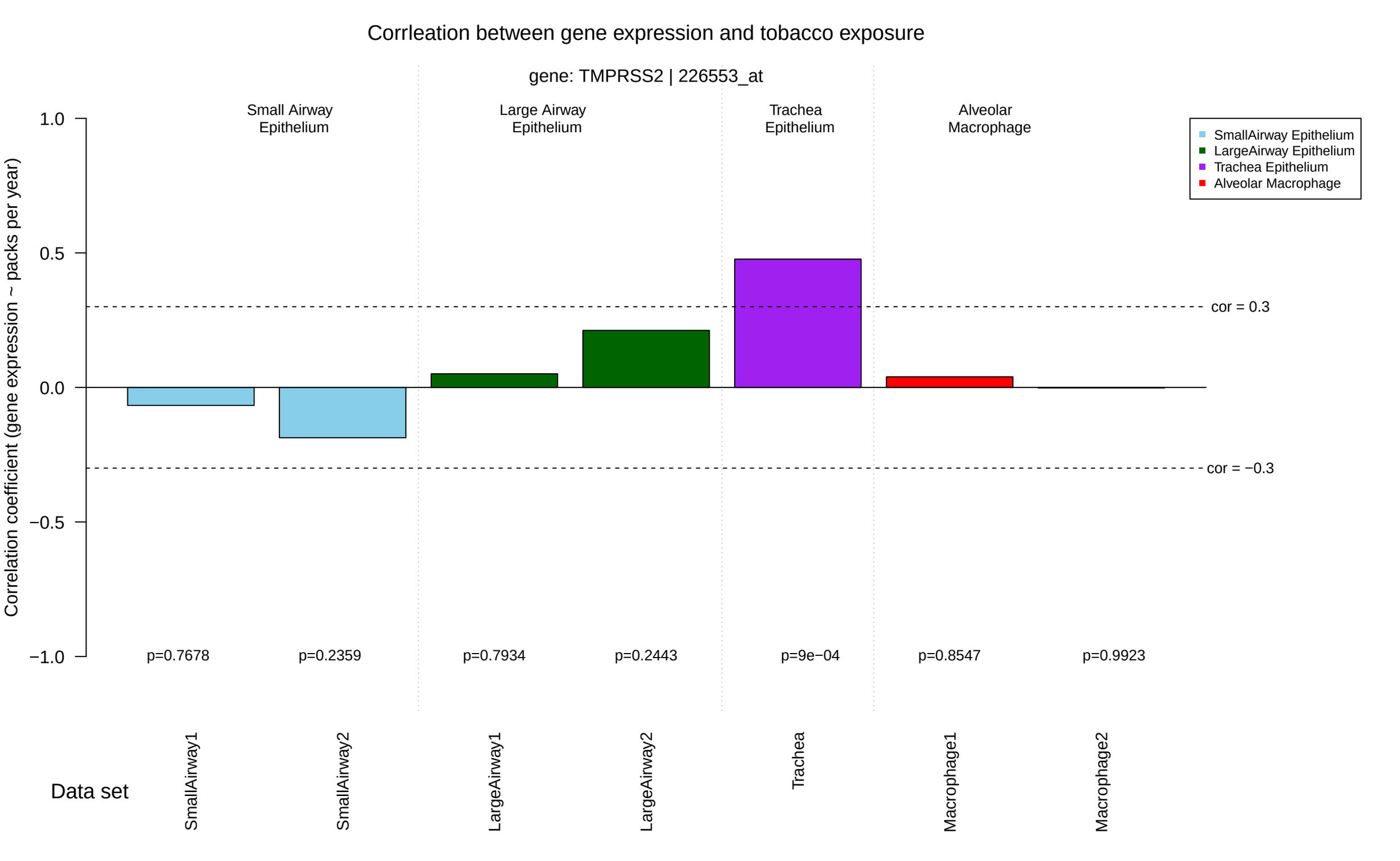

set: LargeAirway1 (p=0.6138)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1878)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8974)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5779)
TMPRSS2 | 226553_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS2 | 226553_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7678
p=0.2359
p=0.7934
p=0.2443
p=9e−04
p=0.8547
p=0.9923
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2